

Programming Problems

1. Write an abstract class `Vehicle`. Derive three classes `Car`, `Motorcycle` and `Truck` from it. Define appropriate methods and print the details of the vehicle.
2. Define a class `Employee`. Display the personal and salary details of five employees using single inheritance.
3. Define a class `Student` with data members as `rollno` and `name`. Derive a class `Fees` from `Student` that has a data member `fees` and functions to submit fees and generate receipt. Derive another class `Result` from `Student` that displays the marks and grade obtained by the student.
4. Define a class `Employee` with data members as `empno`, `name`, and `designation`. Derive a class `Qualification` from `Employee` that has data members `UG`, `PG`, and `experience`. Create another class `Salary` which is derived from both these classes to display the details of the employee and compute their increments based on their experience and educational qualification.
5. Write a program that has a class `Student` to store the details of students in a class. Derive another class `Toppers` from the `Student` that stores records of only top three students of the class.
6. Write a program that has a class `Person`. Derive a class `Baseball_Player` from `Person` and display all the details of a famous baseball player.
7. Write a program that extends the class `Shape` to calculate the area of a circle and a cone. (*Hint: To calculate area of a circle only one variable is required so when creating object, pass the other variable with value 1*)
8. Write a program that extends the class `Result` so that the final result of the `Student` is evaluated based on the marks obtained in tests, activities, and sports.
9. Write a program that extends the `Employee` class so that it stores two more data members `DOB` and `Date of Hiring`. The `Date` must be defined as a separate class.
10. Write a program that has a class `Train` with data members—`no_of_seats_1st`, `no_of_seats_2Tier`, and `no_of_seats_3Tier`—and member functions to set and display data. Derive a class `Reservation` that has data members—`seats_booked_1st`, `seats_booked_2Tier`, and `seats_booked_3Tier`—, and functions to book and cancel tickets, and display status.
11. Write a program that has a class `Distance` with members-kms and metres. Derive classes `School` and `Office` which store the distance from your house to school and office along with other details.
12. Write a program that extends the class `Employee`. Derive a class `Manager` from `Employee` so that it lists all the details of the manager as well as the details of employees working under that manager.
13. Write a program for a publishing company that markets both printed books and audio-visual lectures stored on CDs. Write a class `Publication` that stores

title and price. Derive a class `Book` which has an additional member as `no_pages` and a class `Lecture` with member `play_time`.